

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B. Tech. (EE) Examination

May 2014

EE309 ~~Electrical~~ Power Utilization & Traction

Date: 13.05.2014, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain plugging for D.C. series & shunt motor. [07]

Q - 2 (a) Explain two types of Rotor side Speed control of 3-phases induction motor. [07]

(1) Cascade control. (2) Rotor rheostat control

(b) Explain mechanics of train movement [07]

OR

Q - 2 (a) Explain Rheostatic braking for three phase induction motor. [07]

(b) Draw typical speed time curve for main line service and Calculate V_m by trapezoidal speed time curve. [07]

Q - 3 (a) Explain D.C. and A.C. system of track electrification. [07]

(b) The speed time curve of a train consists of: (i) Uniform acceleration of 6 kmphs for 25 seconds; (ii) Free running for 10 minutes; (iii) Uniform deceleration of 6 kmphs to stop the train; (iv) A stop of 5 minutes; Find the distance between the stations, the average and schedule speeds. [07]

OR

Q - 3 (a) Explain types of electric drives & write feature of an ideal traction system. [07]

(b) A certain motor has to perform the following duty cycle: [02]

100 h.p. for 10 minutes, No load for 5 minutes, 50 h.p. for 8 minutes, No load for 4 minutes. The duty cycle is repeated indefinitely. Draw the curve for the load cycle. Assuming that the heating is proportional to square of the load, determine the suitable size of a continuously rated motor.

(c) An electric train has an average speed of 42 Km/hour on level track between stops 1,400 m apart. It is accelerated at 1.7 kmphs and is braked at 3.3 kmphs. Draw the speed-time curve for the run. [05]

SECTION – II

Q - 4 (a) Derive flywheel calculation equation when motor load is increasing and motor load is decreasing. [07]

Q - 5 (a) Write the Comparison between tungsten Filament lamps and Fluorescent tubes. [07]

(b) State different types of heating methods and explain types of arc furnaces. [07]

OR

Q - 5 (a) Explain spot welding. [06]

(b) Explain types of lighting schemes. [04]

(c) Explain faraday's laws of electro deposition. [04]

Q - 6 (a) Explain dielectric heating. [07]

(b) A 4 pole, 50Hz induction motor has a flywheel on its shaft. Total inertia at the motor shaft is 1000 kg-m^2 . The load torque is 100 kg-m for 10 second followed by a no load period long enough for the flywheel to regain its full speed. Motor has a slip of 6% at a torque of 50 kg-m . Calculate the speed at the end of deceleration period. Assume motor speed-torque characteristic to be a straight line in the region of interest & neglect friction and windage loss. [07]

OR

Q - 6 (a) Explain photometry and list different types of photometer heads. [07]

(b) Explain factory lighting. [07]
